Arhitecturi si Prelucrari Paralele
APP

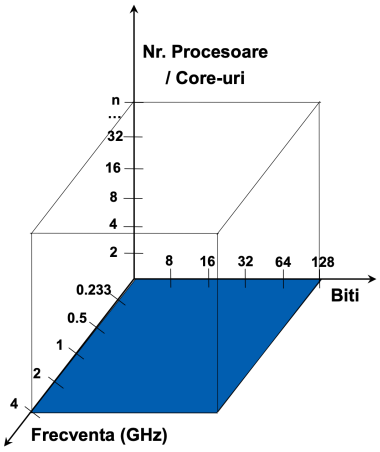
Nicolae Tapus nicolae.tapus@cs.pub.ro
Emil Slusanschi emil.slusanschi@cs.pub.ro

University Politehnica of Bucharest

Computer Science & Engineering Department

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Structuri de Calcul cu Prelucrare Paralela



- *Aria → puterea de calcul*
- *Marimea puterii de calcul: volumul cubului - "n" e limitat "doar" de cost →* ***Arhitecturi paralele de calcul***
- *Orice problema are un grad de paralelism intrinsec ce depinde de natura ei*
- *Trebuie tinut seama de:*
  - *Algoritmi paraleli specifici*
  - *Limbaje adecvate de programare*
  - *Limitarile SO*
  - *Arhitectura intrinseca a SC utilizat*

Gradul general de paralelism al codurilor este ~ 10%

2

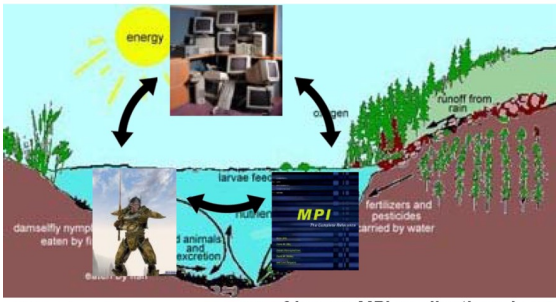
Trends in Parallel Computing in 2010 – 202x+

3

- Computing is changing more rapidly than ever before, and scientists have the unprecedented opportunity to change computing directions

Commercial Off The Shelf technology (COTS)

Supercomputing Ecosystem 20xx



"Clusters" years of legacy MPI applications base

©Horst Simon
LBNL@UC Berkeley

3

Traditional Sources of Performance Improvement are Flat-Lining

4

- **New Constraints**
  - 15 years of *exponential* clock rate growth has ended
- **Moore's Law reinterpreted:**
  - How do we use all of those transistors to keep performance increasing at historical rates?
  - Industry Response: #cores per chip doubles every 18 months *instead* of clock frequency!

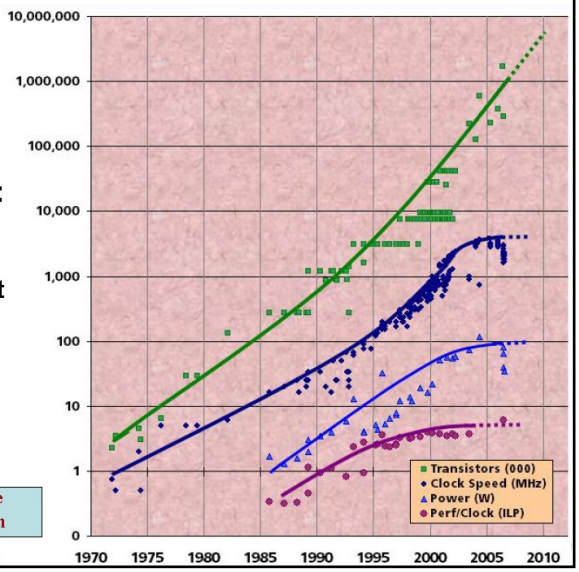


Figure courtesy of Kunle Olukotun, Lance Hammond, Herb Sutter, and Burton Smith

4

5

Supercomputing Ecosystem 20xx

Commercial Off The Shelf technology (COTS)



PCs and desktop systems are no longer the economic driver.





Architecture and programming model are about to change

"Clusters"

years of legacy MPI applications base

5

6

Breaking the 1PFlop Barrier

- 1,026 Tflop/s on LINPACK reported on June 9, 2008
- 6,948 dual core Opteron + 12,960 cell BE
- 80 TByte of memory
- IBM built, installed at LANL



Roadrunner

6

7

Cray XT5@ORNL



Jaguar	Total	XT5	XT4
Peak Performance	1,645	1,382	263
AMD Opteron Cores	**181,504**	**150,176**	**31,328**
System Memory (TB)	**362**	**300**	**62**
Disk Bandwidth (GB/s)	**284**	**240**	**44**
Disk Space (TB)	**10,750**	**10,000**	**750**
Interconnect Bandwidth (TB/s)	**532**	**374**	**157**

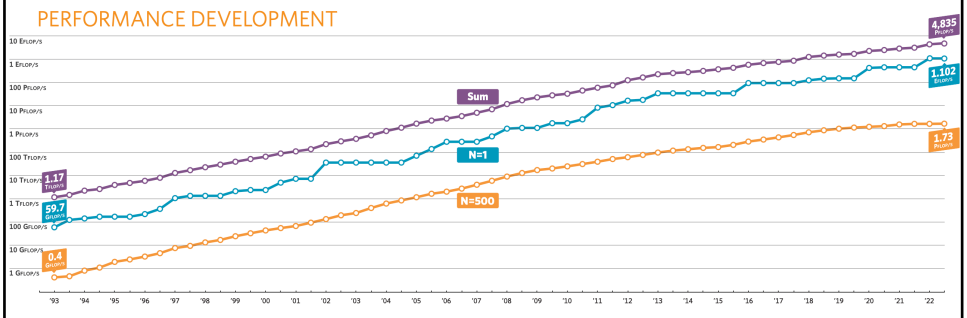
The systems will be combined after acceptance of the new XT5 upgrade. Each system will be linked to the file system through 4x-DDR Infiniband

U.S. DEPARTMENT OF ENERGY
Office of Science

7

8

Top 500 – 1993-2023



8

Moore's Law "Reloaded"

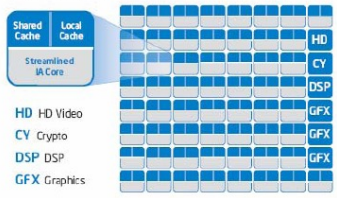
- Number of cores per chip will double every two years
- Clock speed will not increase (possibly decrease)
- Need to deal with systems with millions of concurrent threads
- Need to deal with inter-chip parallelism as well as intra-chip parallelism



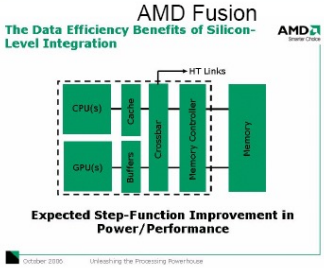
9

Industry presentations show changing trends in processors

Intel's Microprocessor Research Lab



AMD Fusion



Intel's Visual Computing Group - Larabee



nVidia G80 - 2006



Taken from publicly available information

Los Alamos NATIONAL LABORATORY EST. 1943

Operated by Los Alamos National Security, LLC for NNSA

UNCLASSIFIED

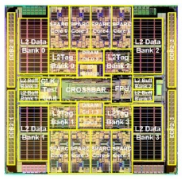
WEAPONS SCIENCE & ENGINEERING CAPABILITY REVIEW

NNSA

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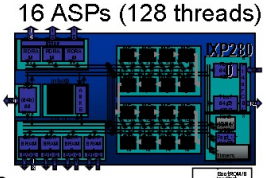
Multicore comes in a wide variety

- Multiple parallel general-purpose processors (GPPs)
- Multiple application-specific processors (ASPs)

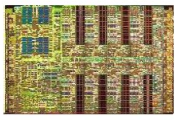


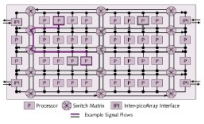
Sun Niagara
8 GPP cores (32 threads)

Intel Network Processor
1 GPP Core
16 ASPs (128 threads)



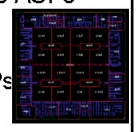
IBM Cell
1 GPP (2 threads)
8 ASPs





Picochip DSP
1 GPP core
248 ASPs

Cisco CRS-1
188 Tensilica GPPs



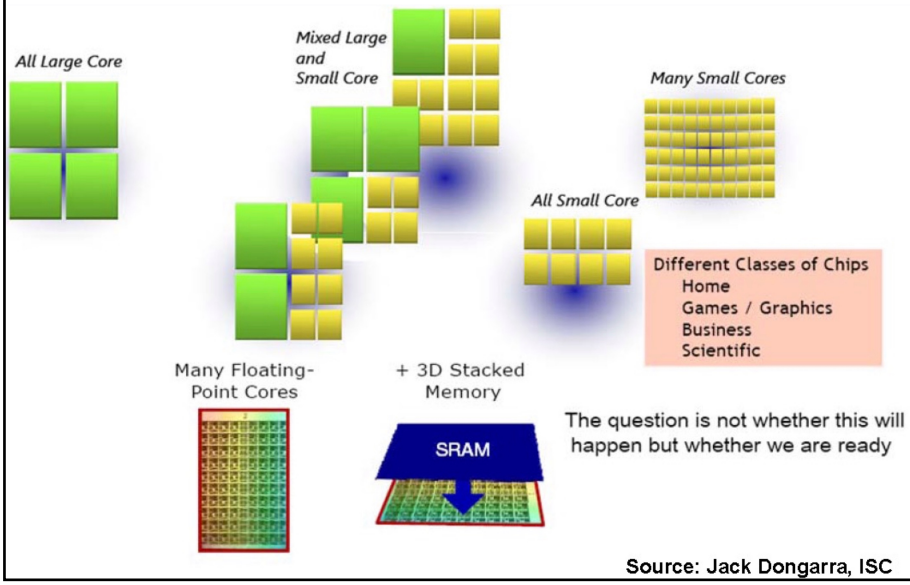


Intel 4004 (1971):
4-bit processor,
2312 transistors,
~100 KIPS,
10 micron PMOS,
11 mm² chip

1000s of processor cores per die

"The Processor is the new Transistor" [Rowen]

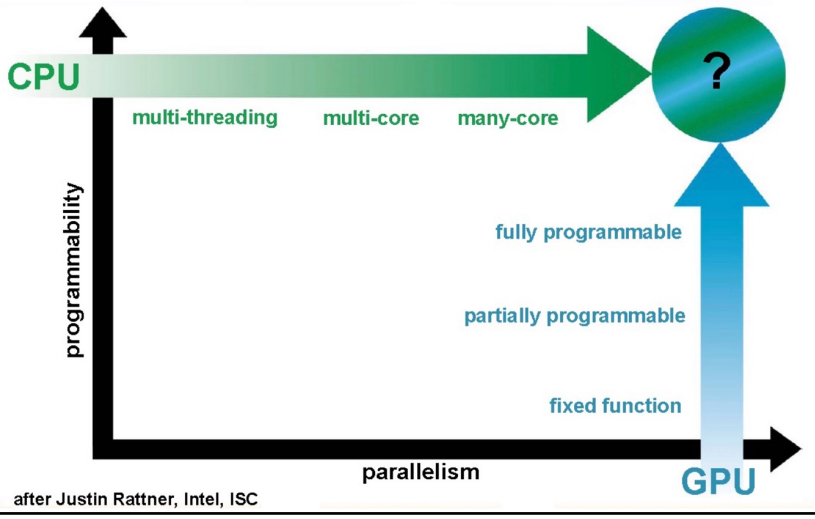
What's next?



Source: Jack Dongarra, ISC

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A Likely Trajectory: Collision or Convergence?



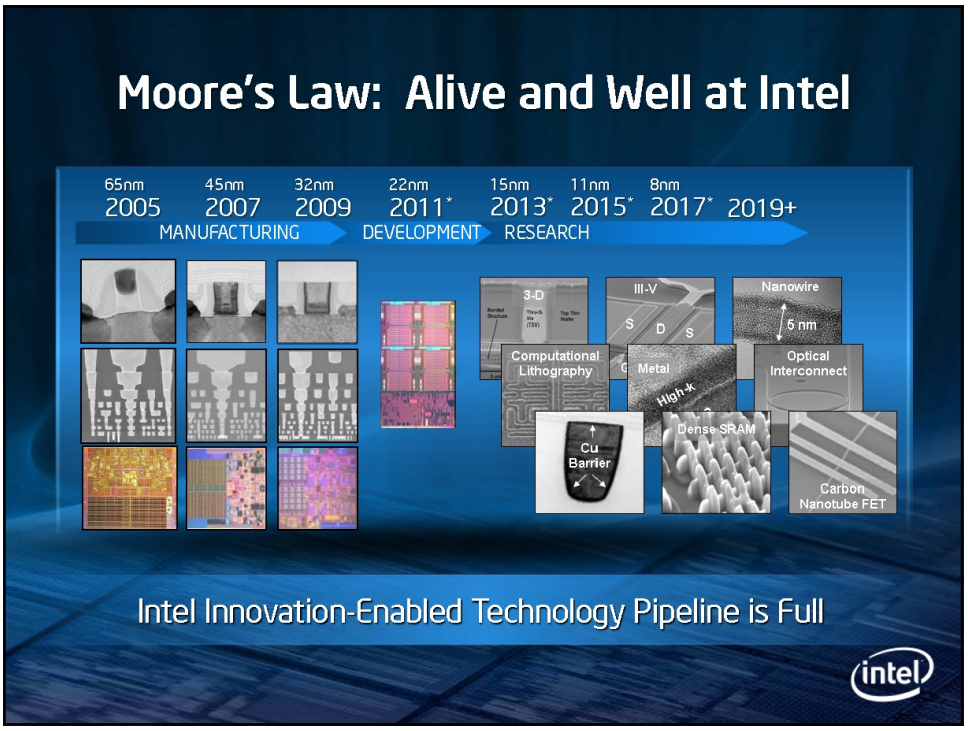
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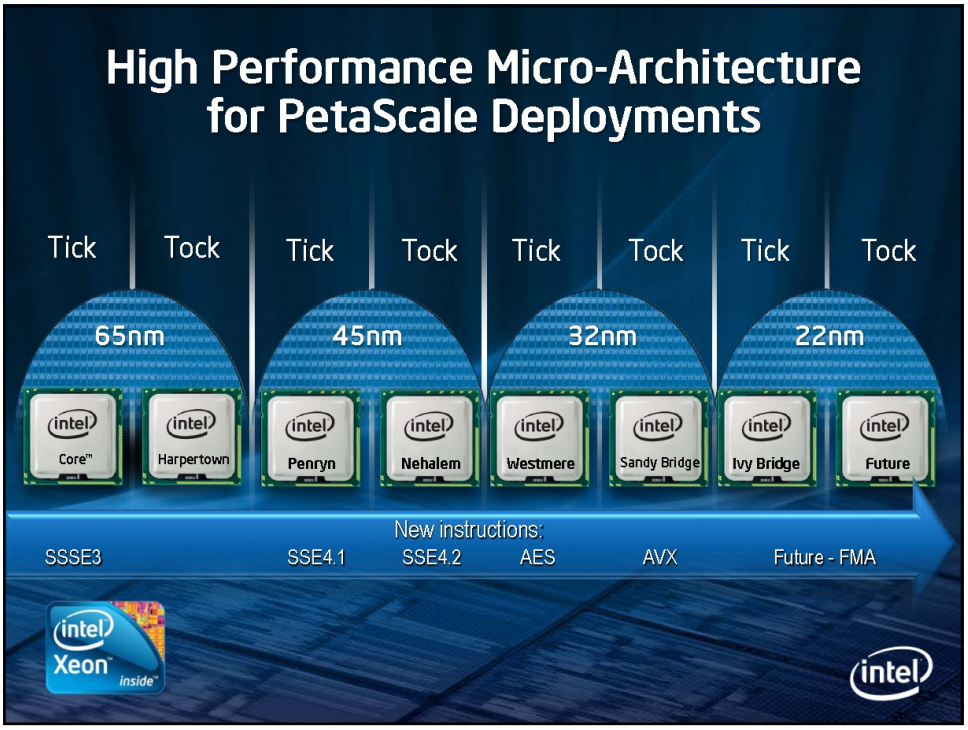
Trends for the next 5-10 years

- After a period of rapid architectural change we will likely settle on a future standard processor architecture
- A good bet:
  - Intel will continue to be an important actor
  - AMD will be the challenger in the CPU space
  - NVidia will lead in the accelerator space
- The impact of this disruptive change on software and systems architecture is not clear yet

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Moore's Law: Alive and Well at Intel
65nm 2005
45nm 2007
32nm 2009
22nm 2011*
15nm 2013*
11nm 2015*
8nm 2017*
2019+
MANUFACTURING
DEVELOPMENT
RESEARCH
3-D
III-V
Nanowire
5 nm
Computational Lithography
Metal
High-k
Optical Interconnect
Cu Barrier
Dense SRAM
Carbon Nanotube FET
Intel Innovation-Enabled Technology Pipeline is Full
intel

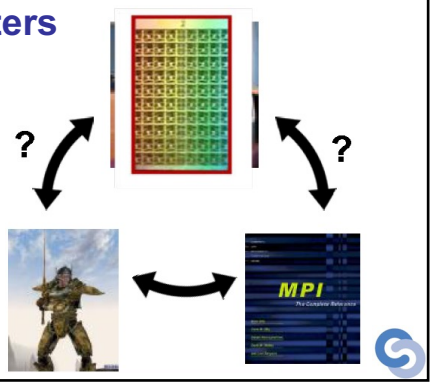
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High Performance Micro-Architecture for PetaScale Deployments
Tick
Tock
65nm
45nm
32nm
22nm
Core™
Harpertown
Penryn
Nehalem
Westmere
Sandy Bridge
Ivy Bridge
Future
New instructions:
SSSE3
SSE4.1
SSE4.2
AES
AVX
Future - FMA
intel Xeon inside™
intel

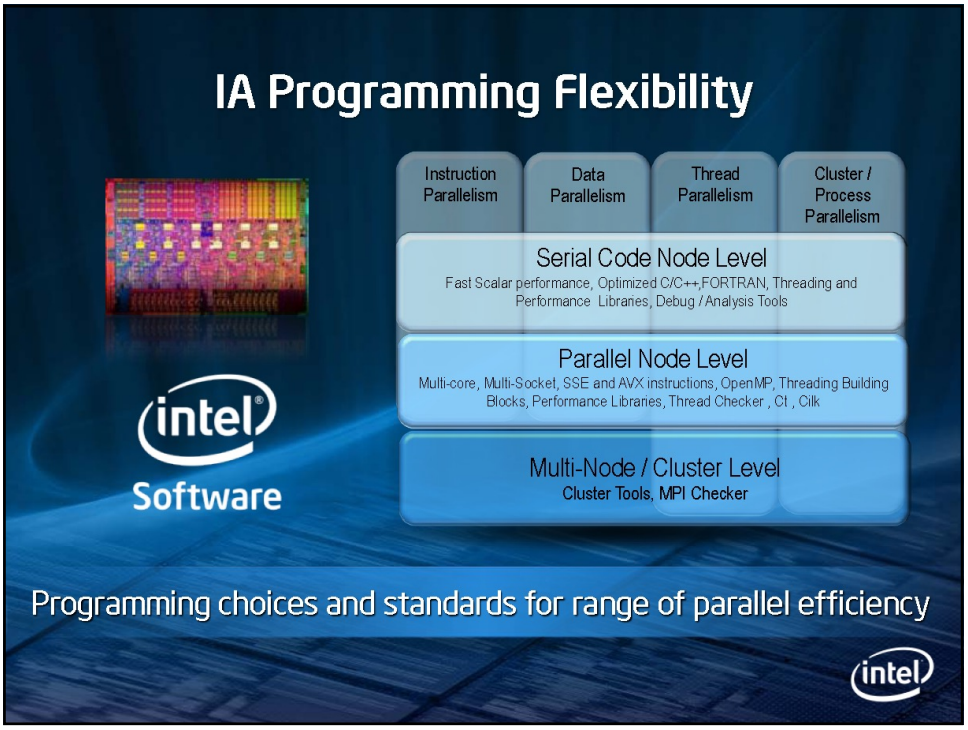
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Impact on Software

- **We will need to rethink and redesign our software**
  - **Similar challenge as the 90's transition to clusters and MPI**
- What will be the ?

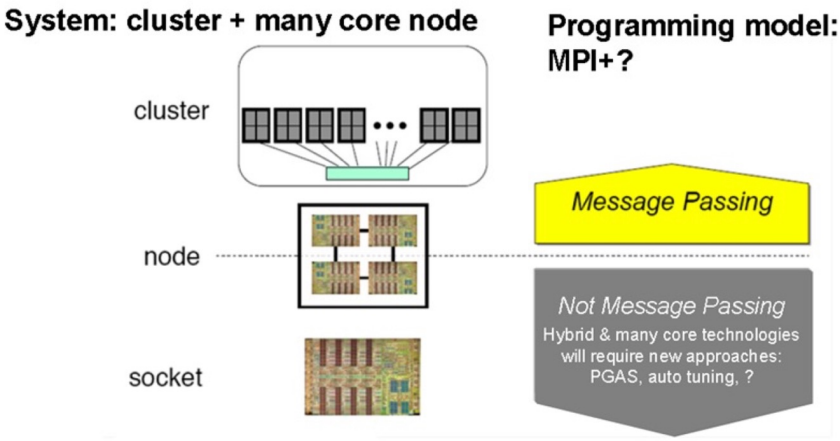


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A Likely Future Scenario



after Don Grice, IBM, Roadrunner Presentation, ISC



Why MPI will persist

- Obviously MPI will not disappear in five years
- Today we have more than 30 years of legacy software in MPI
- New systems are not sufficiently different to lead to a radical new programming model



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What will be the "?" in MPI+?

- Likely candidates are
  - PGAS languages
  - Autotuning
  - CUDA, OpenCL, OpenACC
  - A wildcard from the commercial space



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What's Wrong with MPI Everywhere?

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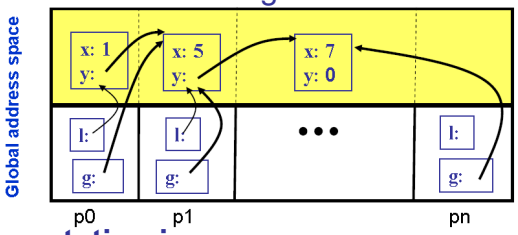
- One MPI process per core is wasteful of intra-chip latency and bandwidth
- **Weak scaling**: success model for the "cluster era" not enough memory per core
- **Heterogeneity**: MPI per CUDA thread-block?



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PGAS Languages

- **Global address space**: thread may directly read/write remote data
- **Partitioned**: data is designated as local or global

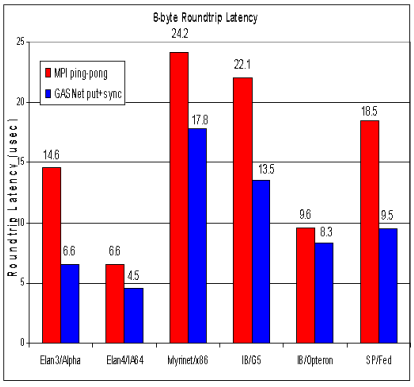


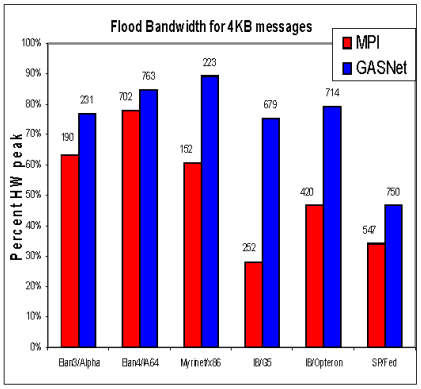
- **Implementation issues:**
  - Distributed memory: Reading a remote array or structure is explicit, not a cache fill
  - Shared memory: Caches are allowed, but not required
- No less scalable than MPI!
- Permits sharing, whereas MPI rules it out!



Performance Advantage of One-Sided Communication

- The put/get operations in PGAS languages (remote read/write) are one-sided (no required interaction from remote proc)
- This is faster for pure data transfers than two-sided send/receive



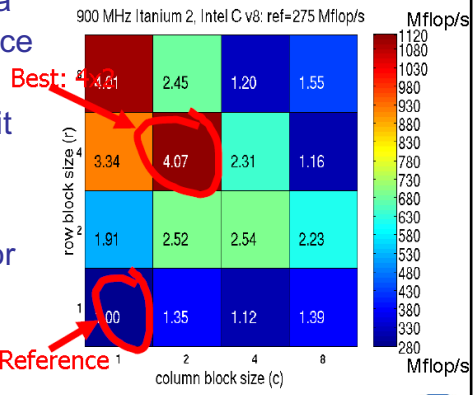




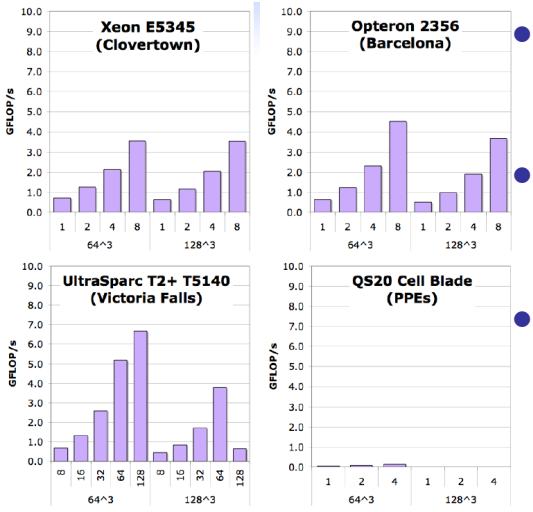
Autotuning

- Write programs that write programs
  - Automate search across a complex optimization space
  - Generate space of implementations, search it
  - Performance far beyond current compilers
  - Performance portability for diverse architectures!
  - Past success stories: PhiPAC, ATLAS, FFTW, Spiral, OSKI

Finite Element Problem
[Im, Yelick, Vuduc]



Performance

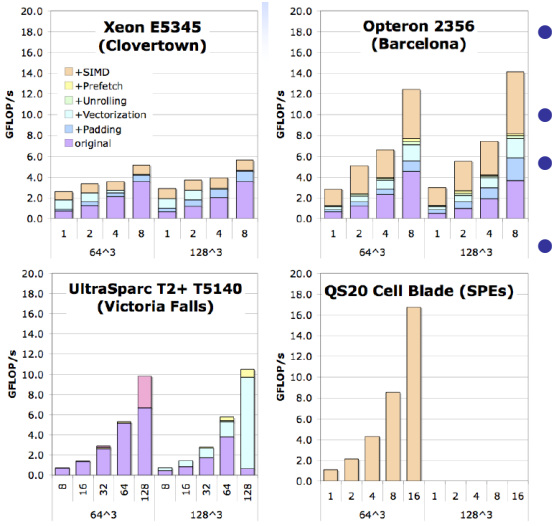


- Reference code was nominated for a Gordon Bell prize
- Used for out-of-box study on multicore performance
- Superficially, scalability looks good, but is performance good?
  - **No**
  - Performance model

POC: Leonid Oliker, Samuel Williams (LBNL)

Reference+NUMA

Autotuning



- Auto-tuning dramatically improved performance
- Clearly no silver bullet
- SIMD partially breaks portability premise
- Note: Cell version was optimized, not auto-tuned

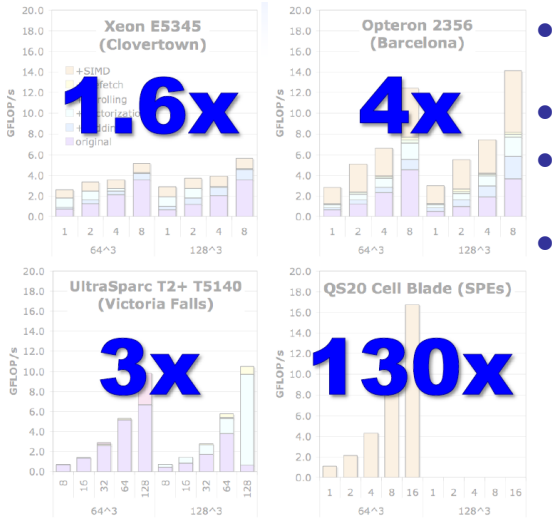


POC: Leonid Oliker, Samuel Williams (LBNL)



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Autotuning



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POC: Leonid Oliker, Samuel Williams (LBNL)



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